Elementary Analysis

Section 25: More on Uniform Convergence

Exercises 25.1–25.15, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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25 More on uniform convergence

Three useful tools. A uniform error bound can be integrated on a finite interval: $|\int_a^b (f_n - f)| \le (b-a)\sup|f_n - f|$. A series of functions converges uniformly precisely when its partial sums do. Finally, the Weierstrass M-test says that $|g_k(x)| \le M_k$ for all inputs, together with $\sum M_k < \infty$, guarantees uniform convergence of $\sum g_k$. If all summands are continuous, its sum is continuous too. An M-test bound must not depend on the input x.

25.1 Exercise 25.1: Bounding an integral by the integral of an absolute value

What we must show. Deduce $|\int_a^b g| \le \int_a^b |g|$ from monotonicity of the integral. The integrals in the statement are assumed to exist.

Place the function between two integrable bounds.

For every x, $-|g(x)| \le g(x) \le |g(x)|$. Applying monotonicity first to the lower pair and then to the upper pair gives

$$-\int_a^b |g(x)|\,dx \le \int_a^b g(x)\,dx \le \int_a^b |g(x)|\,dx.$$

Write the middle integral as A and the nonnegative right-hand integral as B. The statement $-B \le A \le B$ is exactly the statement $|A| \le B$.

Conclusion. $\left|\int_a^b g(x)\,dx\right| \le \int_a^b |g(x)|\,dx.$

25.2 Exercise 25.2: Powers divided by their indices

What we must show. Show uniform convergence of $f_n(x) = x^n/n$ on $[-1, 1]$.

Find a common error bound.

Since $|x| \le 1$, $|x^n/n| \le 1/n$ for every input in this interval. Therefore the pointwise limit is $f(x) = 0$. Given $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. Then $n > N$ implies $|f_n(x) - 0| \le 1/n < \varepsilon$ for *all* $x \in [-1, 1]$ simultaneously.

Conclusion. The convergence is uniform and the limit function is identically zero.

25.3 Exercise 25.3: A rational trigonometric expression and its integral

What we must show. Study $f_n(x) = (n + \cos x)/(2n + \sin^2 x)$ on $\mathbb{R}$.

(a) Guess the limit, then calculate its exact error.

Dividing by n suggests the limit $1/2$. Subtract it with a common denominator:

$$f_n(x) - \frac{1}{2} = \frac{2\cos x - \sin^2 x}{2(2n + \sin^2 x)}.$$

Since $|2\cos x - \sin^2 x| \le 2 + 1 = 3$ and the denominator is at least $4n$,

$$\left|f_n(x) - \frac{1}{2}\right| \le \frac{3}{4n} \qquad (x \in \mathbb{R}).$$

The right-hand side tends to zero independently of x, proving uniform convergence.

(b) Apply the integral theorem after checking its hypotheses.

The denominator is positive, so every f_n is continuous on $[2, 7]$. The convergence is uniform on that interval by part (a). Theorem 25.2 therefore gives

$$\lim_{n\to\infty} \int_2^7 f_n(x)\,dx = \int_2^7 \frac{1}{2}\,dx = \frac{5}{2}.$$

Conclusion. The uniform limit on $\mathbb{R}$ is $1/2$, and the integral limit is $5/2$.

25.4 Exercise 25.4: Uniform convergence implies the uniform Cauchy property

What we must show. Show that $f_n \to f$ uniformly on S makes (f_n) uniformly Cauchy.

Use the same approximating limit for two indices.

Given $\varepsilon > 0$, choose N so that $|f_k(x) - f(x)| < \varepsilon/2$ whenever $k > N$ and $x \in S$. For *both* $m > N$ and $n > N$, the triangle inequality then gives

$$|f_n(x) - f_m(x)| \le |f_n(x) - f(x)| + |f(x) - f_m(x)| < \varepsilon \qquad (x \in S).$$

The same N works for every pair of sufficiently large indices and every input, which is exactly uniform Cauchyness.

Conclusion. Every uniformly convergent sequence is uniformly Cauchy.

25.5 Exercise 25.5: A uniform limit of bounded functions is bounded

What we must show. If all f_n are bounded on S and $f_n \to f$ uniformly, prove that f is bounded.

Plan. The bounds of all the f_n need not be the same. One suitably chosen approximating function is enough.

Choose one function within distance one of the limit.

Uniform convergence gives an index m such that $|f(x) - f_m(x)| < 1$ for all $x \in S$. Since f_m is bounded, some finite M satisfies $|f_m(x)| \le M$ for every $x \in S$.

Add the approximation error to that fixed bound.

For every input,

$$|f(x)| \le |f(x) - f_m(x)| + |f_m(x)| < 1 + M.$$

This single finite bound establishes boundedness of f.

Conclusion. f is bounded; a bound is $M + 1$ for the chosen approximant.

25.6 Exercise 25.6: Absolute summability gives continuity at the endpoints too

What we must show. Relate $\sum |a_k| < \infty$ to the power series on $[-1, 1]$.

(a) Apply the M-test on the full closed interval.

For every $x \in [-1,1]$, $|a_k x^k| \le |a_k|$. The numerical majorant series $\sum |a_k|$ converges by assumption. The M-test therefore gives uniform convergence of $\sum a_k x^k$ on $[-1,1]$. Every partial sum is a polynomial and hence continuous. Its uniform limit is continuous on $[-1,1]$, including the endpoints with the relative notion of continuity.

(b) Check the coefficient condition in the given example.

For $a_n = 1/n^2$, $n \ge 1$, the numerical series $\sum 1/n^2$ converges by the p-series test, since $2 > 1$. Part (a) therefore applies directly to $\sum_{n=1}^{\infty} x^n/n^2$.

Conclusion. Both the general series and the specified example have continuous sums on the whole closed interval $[-1,1]$.

25.7 Exercise 25.7: A uniformly convergent trigonometric series

What we must show. Prove that $\sum_{n=1}^{\infty} \cos(nx)/n^2$ defines a continuous function on $\mathbb{R}$.

Bound every summand on the unbounded domain.

The elementary estimate $|\cos(nx)| \le 1$ gives $|\cos(nx)/n^2| \le 1/n^2$ for all real x. Since $\sum 1/n^2$ converges, the M-test proves uniform convergence on $\mathbb{R}$.

Transfer continuity from finite sums to the limit.

Each summand is continuous, so every partial sum is continuous. Theorem 24.3 applied to this uniformly convergent sequence of partial sums makes the sum continuous on $\mathbb{R}$.

Conclusion. The series converges uniformly on $\mathbb{R}$ to a continuous function.

25.8 Exercise 25.8: A power series uniform on its entire convergence interval

What we must show. Find the radius and endpoint behavior of $\sum_{n=1}^{\infty} x^n/(n^2 2^n)$.

Compute the radius using coefficient roots.

With $a_n = 1/(n^2 2^n)$,

$$|a_n|^{1/n} = \frac{1}{2n^{2/n}} \longrightarrow \frac{1}{2},$$

since $n^{1/n} \to 1$. Thus the radius is $R = 1/(1/2) = 2$.

Bound the terms at all inputs, including the two endpoints.

For $|x| \le 2$,

$$\left|\frac{x^n}{n^2 2^n}\right| \le \frac{1}{n^2}.$$

The M-test proves uniform convergence on $[-2,2]$, not merely on smaller intervals. The summands are polynomials, so the sum is continuous there.

Conclusion. The radius is 2, and the series converges uniformly to a continuous function on its full interval of convergence $[-2,2]$.

25.9 Exercise 25.9: Geometric series on a smaller and a maximal open interval

What we must show. Compare uniform convergence of $\sum_{n=0}^{\infty} x^n$ on $[-a, a]$ and $(-1, 1)$, where $0 < a < 1$.

(a) Estimate the geometric remainder on the smaller interval.

For $|x| < 1$, the sum is $1/(1-x)$ and the partial sum through N has remainder

$$\frac{1}{1-x} - \sum_{n=0}^{N} x^n = \frac{x^{N+1}}{1-x}.$$

For $|x| \le a$, its absolute value is at most $a^{N+1}/(1-a)$, which tends to zero because $0 < a < 1$. The estimate is uniform in x.

(b) Show that terms do not even tend uniformly to zero on $(-1, 1)$.

If the partial sums converged uniformly, their successive differences x^N would converge uniformly to zero: indeed each difference is bounded by the two uniform errors of the corresponding partial sums. But at $x_N = 2^{-1/N} \in (0, 1)$, we have $x_N^N = 1/2$ for every $N \ge 1$. Thus the necessary uniform vanishing of the terms fails.

Conclusion. Uniform convergence holds on every $[-a, a]$ with $a < 1$, and fails on $(-1, 1)$. Pointwise convergence on the latter interval still holds.

25.10 Exercise 25.10: A non-geometric series compared with a geometric one

What we must show. Analyze $\sum x^n/(1+x^n)$ on $[0, 1)$. The starting index may be taken as $n = 0$; its term is just the constant $1/2$ and does not affect any convergence decision below.

(a) Prove convergence for a fixed input.

For $0 \le x < 1$ and $n \ge 1$,

$$0 \le \frac{x^n}{1+x^n} \le x^n.$$

The majorant $\sum_{n=1}^{\infty} x^n$ converges, so comparison proves convergence of the given nonnegative series at every such fixed x.

(b) Obtain a majorant independent of the input in $[0, a]$.

If $0 \le x \le a < 1$, then $x^n/(1+x^n) \le a^n$. Since $\sum a^n$ converges, the M-test proves uniform convergence on $[0, a]$.

(c) Test the necessary uniform vanishing of the summands.

Take $x_n = 2^{-1/n} \in (0, 1)$. Then $x_n^n = 1/2$ and

$$\frac{x_n^n}{1+x_n^n} = \frac{1/2}{3/2} = \frac{1}{3}.$$

Thus the terms do not tend uniformly to zero on $[0, 1)$. Uniform convergence of the series would force that property by subtracting successive partial sums, as in Exercise 25.9. Hence uniform convergence fails.

Conclusion. There is pointwise convergence on $[0,1)$ and uniform convergence on every $[0,a]$ with $a<1$, but no uniform convergence on $[0,1)$.

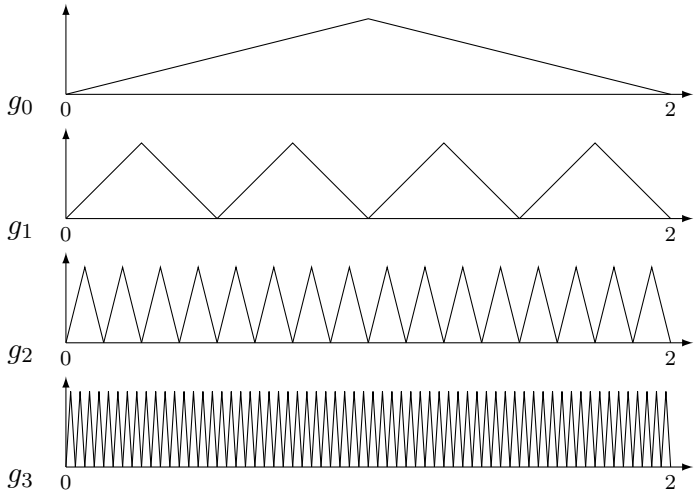
25.11 Exercise 25.11: Four increasingly compressed triangular waves

What we must show. Sketch the $g_n(x) = g(4^n x)$ from Example 3 and prove continuity of $f(x) = \sum_{n=0}^{\infty}(3/4)^n g_n(x)$.

(a) Describe the graphs by exact vertices.

The base function g is the triangular wave of period 2: for every integer j, $g(2j)=0$, $g(2j+1)=1$, and its graph is straight between consecutive integer inputs. Thus g_n has zeros at $2j/4^n$, peaks of height 1 at $(2j+1)/4^n$, and period $2/4^n$. Multiplication of the input compresses the horizontal scale; it does not change the heights.

The sketches below show $0 \le x \le 2$, with the same horizontal and vertical scales in all four rows; repeat each graph periodically in both directions. For g_3, the closely spaced peaks are intentional.



(b) Bound the vertically weighted waves.

The triangular wave is continuous, with $0 \le g \le 1$. Thus every g_n is continuous as the composition of g with the continuous map $x \mapsto 4^n x$, and

$$\left|\left(\frac{3}{4}\right)^n g_n(x)\right| \le \left(\frac{3}{4}\right)^n \qquad (x \in \mathbb{R}).$$

The numerical geometric series $\sum_{n=0}^{\infty}(3/4)^n$ converges. The M-test gives uniform convergence on $\mathbb{R}$, and the uniform-limit theorem gives continuity of f on $\mathbb{R}$.

Conclusion. The graphs have periods 2, 1/2, 1/8, 1/32 and height 1. The weighted infinite sum is continuous everywhere. No differentiability claim is needed for this exercise.

25.12 Exercise 25.12: Integrating a uniformly convergent series

What we must show. If continuous g_k have a uniformly convergent sum g on $[a,b]$, prove $\int_a^b g = \sum_{k=1}^{\infty}\int_a^b g_k$.

Apply the known theorem to partial sums, not directly to infinite sums.

Set $s_n(x) = \sum_{k=1}^{n} g_k(x)$. Each s_n is continuous, and the hypothesis means $s_n \to g$ uniformly. Theorem 25.2 therefore gives $\int_a^b s_n \to \int_a^b g$.

Use linearity only for the finite sum, then take its limit.

For every n,

$$\int_a^b s_n(x)\,dx = \sum_{k=1}^{n} \int_a^b g_k(x)\,dx.$$

Consequently the numerical series on the right converges, and its sum is $\int_a^b g(x)\,dx$ by the definition of an infinite series as a limit of finite partial sums.

Conclusion. Term-by-term integration is valid under the stated uniform convergence and continuity assumptions.

25.13 Exercise 25.13: Adding two uniformly convergent series

What we must show. Show that $\sum(g_k + h_k)$ converges uniformly if $\sum g_k$ and $\sum h_k$ do.

Name the two limits and their partial sums.

Write $G_n = \sum_{k=1}^{n} g_k \to G$ uniformly and $H_n = \sum_{k=1}^{n} h_k \to H$ uniformly. The partial sum of the proposed series is exactly $G_n + H_n$.

Estimate the error of that partial sum.

For every input x,

$$|(G_n + H_n)(x) - (G + H)(x)| \le |G_n(x) - G(x)| + |H_n(x) - H(x)|.$$

Given $\varepsilon > 0$, make each error smaller than $\varepsilon/2$ uniformly, and take the larger of the two index thresholds. This gives uniform convergence to $G + H$.

Conclusion. The sum series is uniformly convergent, with sum equal to the sum of the two original sum functions.

25.14 Exercise 25.14: Multiplying a series by one bounded function

What we must show. If $\sum g_k$ converges uniformly on S and h is bounded there, prove uniform convergence of $\sum h g_k$.

Write the new partial sums as a product with the old ones.

Let $G_n = \sum_{k=1}^{n} g_k \to G$ uniformly and choose a finite $M \ge 0$ with $|h(x)| \le M$ for all $x \in S$. Then

$$\sum_{k=1}^{n} h(x)g_k(x) = h(x)G_n(x), \qquad |h(x)G_n(x) - h(x)G(x)| \le M|G_n(x) - G(x)|.$$

Choose the uniform error tolerance after accounting for the bound.

If $M = 0$, every new term is zero and there is nothing to prove. If $M > 0$, choose N so that $|G_n(x) - G(x)| < \varepsilon/M$ for every $x \in S$ and $n > N$. The displayed inequality then bounds the new error by ε, uniformly in x.

Conclusion. The new series converges uniformly to hG. No continuity of h is required, only its boundedness.

25.15 Exercise 25.15: Monotone convergence on a compact interval: Dini's theorem

What we must show. Prove both monotone versions of the uniform-convergence assertion.

(a) First note that decreasing terms converging to zero are nonnegative.

For each fixed x, $f_n(x) \geq \lim_m f_m(x) = 0$. We must show that the nonnegative errors $f_n(x)$ become small at one common index.

Use continuity to turn each strict error bound into an open set.

Fix $\varepsilon > 0$, and let

$$U_n = \{x \in [a, b] : f_n(x) < \varepsilon\}.$$

Continuity makes U_n relatively open in $[a, b]$. Since $f_{n+1} \leq f_n$, these sets increase: $U_n \subseteq U_{n+1}$. Pointwise convergence to zero implies every x lies in some U_n, so $\{U_n : n \geq 1\}$ is an open cover of $[a, b]$.

Compactness turns the varying pointwise indices into one common index.

The closed bounded interval is compact, so finitely many of the U_n cover it. Let N be the largest of their indices. Because the sets are nested, U_N alone covers the whole interval. Thus $0 \leq f_n(x) \leq f_N(x) < \varepsilon$ for all $x \in [a, b]$ and $n \geq N$. This is uniform convergence to zero.

(b) Reduce increasing convergence to the decreasing case.

Suppose now $f_n(x)$ increases to a continuous function $f(x)$. Set $h_n = f - f_n$. Each h_n is continuous because both f and f_n are. At every input h_n decreases to zero, and $h_n \geq 0$. Part (a) gives $h_n \to 0$ uniformly, which is exactly $f_n \to f$ uniformly.

Conclusion. Both claims hold. Compactness and continuity of the limiting function are essential ingredients in turning the pointwise monotonicity into a uniform bound.